PRAVINKUMAR GHODAKE

PRAVINKUMAR RAMCHANDRA GHODAKE

Digital Footprint & Public Records

Section Description: This section provides a structured repository of independently verifiable professional, academic, research, institutional, publication, and documentary records associated with Pravinkumar Ghodake. Each external reference is included because it is directly relevant to establishing identity, authorship, affiliation, research activity, publication history, or documented chronology.

The references below are organized by source type and documentary function. Priority is given to institutional records, publisher records, professional organizations, research identifiers, conference records, and other independently accessible sources. Descriptions are limited to information supported by the corresponding source.

Source categories: Institutional · Publisher · Professional Society · Research Identifier · Conference · Professional Profile · Documentary Archive

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🎓 Academic & Research Profiles

- **Google Scholar:** Research publication and citation profile.

  Source: https://scholar.google.com/citations?user=WJSL6fIAAAAJ&hl=en&oi=ao

- **ResearchGate:** Research profile containing publications, research interests, and academic affiliations.

  Source: https://www.researchgate.net/profile/Pravinkumar_Ghodake

- **UK Metamaterials Network:** Official profile detailing expertise in applied mechanics, nonlinear ultrasonics, and wave scattering.

  Source: https://metamaterials.network/user/pravinkumar-ghodake/

- **ORCID ID:** ORCID — iD 0000-0002-6755-4003. Persistent identifier for research outputs.

  Source: https://orcid.org/0000-0002-6755-4003

- **SPIE Profile:** Official profile within the international society for optics and photonics.

  Source: https://spie.org/profile/Pravinkumar.Ghodake-4330088?SSO=1

- **iMechanica:** iMechanica User Profile – Academic portfolio and contributions to the web of mechanics and mechanicians.

  Source: https://imechanica.org/user/10007432

💼 External Research & Documentary Records

Section Description: This subsection contains selected external research, publication, conference, repository, and documentary records associated with Pravinkumar Ghodake. The sources provide independently accessible records of research activity, publications, presentations, datasets, and related academic work.

- **NDT.net Open Archive (ISNDT Virtual Conference) – Pravinkumar Ghodake:** Shape Optimization for the Design of Linear and Nonlinear Metamaterials to Improve Nonlinear Ultrasonic Testing – Published conference paper on non-gradient algorithms designed to

isolate pure material damage variables from system-generated higher harmonics utilizing integrated COMSOL Multiphysics and MATLAB automated loop architectures.

Source: https://www.ndt.net/search/docs.php?id=26801

- **ResearchGate Publication Hub Technical Poster – Pravinkumar Ghodake & Saurabh Biswas:** Nonlinear One-Way Two-Wave Mixing in a Media with Rate Independent Hysteresis (Additional Data as Poster) – technical poster detailing one-dimensional wave mixing applications across Bouc-Wen and two-states hysteresis models to evaluate sum and difference frequencies for early-stage structural micro-crack quantification.

  Source: https://www.researchgate.net/publication/361282474_Nonlinear_One-Way_Two-Wave_Mixing_in_a_Media_with_Rate_Independent_Hysteresis_Additional_Data_as_Poster

- **The Journal of the Acoustical Society of America Meeting Abstract – Pravinkumar Ghodake:** The complexity of harmonically scattered nonlinear waves from triangular, circular, and rectangular corners of the 2-D domain – Published research abstract under the Acoustical Society of America exploring bulk wave mode conversions, harmonic energy redistribution, and pulse-echo ultrasonic techniques.

  Source: https://pubs.aip.org/asa/jasa/article/154/4_supplement/A262/2924505/The-complexity-of-harmonically-scattered-nonlinear

- **GRC & GRS YouTube Broadcast Archive (Los Angeles & Santa Barbara) – Pravinkumar Ghodake:** Pravinkumar GRC & GRS Memory 16, Los Angeles & Santa Barbara – Video log documenting travel, presentations, and seminar benchmarks at the Gordon Research Seminar & Conference on Multifunctional Materials & Structures in California.

  Source: https://www.youtube.com/watch?v=h3Qv1IBkSDQ

- **IPMS 2022 Conference Minisymposium (Malta) Registry (M8) Presentation – Pravinkumar Ghodake:** Inverse Problems in Science and Engineering – Official program entry from the 10th International Conference on Inverse Problems in Modeling and Simulation (IPMS 2022). Documenting my invited speaker presentation titled *"Inverse problem for quantification of localized damage using 1D nonlinear elastic two-wave mixing"* within the specialized minisymposium co-organized by Ghent University and the University of Malta.

  Source: https://www.ipms-conference.org/ipms2022/index.php/m8

- **ISTAM Proceedings Young Scientist Award Session (IIT Kharagpur) – Pravinkumar Ghodake:** Harmonic Scattering of S0 Lamb Wave - A Computational Study – Award session paper ("Paper for the Young Scientist Award") tracking numerical simulations of nonlinear elastic inclusions in thin plates. It details backscattered higher harmonics under one-way two-wave mixing and draws key physics analogies between harmonically scattered elastic guided waves and 'Raman Scattering' in nonlinear optics.

  Source: https://istam.iitkgp.ac.in/resources/2021/proceedings/Award_session/Abstract/SM/138abstract.pdf

- **NODYCON Open Repository (Metamaterials Inverse Design) – Pravinkumar Ghodake:** Inverse Design of Periodic and Quasi-Periodic Nonlinear Mechanical Metamaterial – Published conference proceeding tracking a universal inverse design strategy implemented to systematically suppress 2^{nd} ($2f$) and 3^{rd} ($3f$) system-generated power amplifier harmonics. This specific research leverages the Nelder-Mead optimization algorithm to adjust individual metallic and glass layer widths within linear, periodic, and quasi-periodic layered structures to seamlessly trap harmonically scattered waves.

  Source: https://nodycon.org/2023/papers/190/abstract_submissions/569/view_abstract

- **NODYCON Open Repository (Crossed-Thin-Rectangular Inclusions) – Pravinkumar Ghodake:** Harmonic Scattering of Waves from Crossed-Thin-Rectangular Nonlinear Inclusions – Published conference proceeding tracking wave manipulation using the finite

element method to model Murnaghan hyperelastic inclusions. It demonstrates the discovery of an "elastically invisible" configuration where 32 periodically spaced, uniquely oriented crossed-rectangular inclusions completely cancel out 2^{nd} ($2f$) and 3^{rd} ($3f$) harmonics alongside static terms ($0f$) through complex interference and energy trapping.

Source: https://nodycon.org/2023/papers/337/abstract_submissions/581/view_abstract

- **JASA Acoustical News Committee Appointment Report – Pravinkumar Ghodake | Acoustical News (AIP Publishing):** 2024–2025 Annual Report of Technical Committees – Official institutional record published in *The Journal of the Acoustical Society of America* documenting the expansion metrics and administrative updates of the Technical Committee on Computational Acoustics (CA TC) under the Acoustical Society of America (ASA).

  Source: https://pubs.aip.org/asa/jasa/article/156/4/2132/3315436/Acoustical-News

- **APS March Meeting Presiders & Session Chair Registry – Pravinkumar Ghodake:** Matter under Extreme Conditions (Session AAA08) – Official record from the 2023 American Physical Society (APS) March Meeting documenting my selection and service as the formal **Session Chair/Presider** over the oral presentation program tracking ab initio workflows, Planck gravity, thermoelastic databases, and machine-learning.

  Source: https://meetings-archive.aps.org/mar/2023/aaa08/

- **ASA Technical Program Calendar (187^{th} Meeting) – Pravinkumar Ghodake:** Exploring Interesting Effects of Symmetric Functionally Graded Unit Cells for the Design of Nonlinear Mechanical Metamaterials – technical presentation abstract published in *The Journal of the Acoustical Society of America*. This research documents a novel functionally graded unit cell that introduces a spatially continuous distribution of nonlinear parameters to mirror 'Raman Scattering' physics and achieve unprecedented, non-intuitive control over forward and backscattered wave harmonics.

Source: https://acousticalsociety.org/wp content/uploads/2024/11/Program_187th_meeting.pdf

- **YouTube Transit Infrastructure Log (Hong Kong Airport Express) – Pravinkumar Ghodake:** Airport Express Train Journey | Hong Kong – Travel log documenting structural transit infrastructure systems and international research connectivity vectors.

  Source: https://www.youtube.com/watch?v=BImYNS7DZSc

- **YouTube Conference Travel Archive (Gordon Research Conference & Gordon Research Seminar (GRC & GRS) Memory Log 68) – Pravinkumar Ghodake:** Pravinkumar GRC & GRS Memory 68, Los Angeles & Santa Barbara – Video log, travel itineraries, and analytical findings from the Gordon Research Conference (GRC) on Multifunctional Materials & Structures in California.

  Source: https://www.youtube.com/watch?v=6zyjLIHOz2w

- **IIT Kanpur Mechanical Engineering Research Day Poster – Pravinkumar Ghodake:** Impact Response of Glass Fiber Composites – Research Day poster presentation completed at the Department of Mechanical Engineering, Indian Institute of Technology Kanpur (IIT Kanpur). Advised under Dr. P. Venkitanarayanan, this research utilizes 3D Digital Image Correlation (3D DIC) coupled with a Hopkinson Pressure Bar (HPB) apparatus and momentum traps to map out-of-plane displacement profiles, back-face delamination propagation, and dynamic local fiber breakage in [0/90/+45/-45] epoxy-glass fiber laminates under drop-weight and high-velocity bullet impacts.

  Source: https://www.researchgate.net/publication/297759952_Impact_Response_of_Glass_Fiber_Composites

- **2nd National Conference on MDAO (Bangalore) Paper – Pravinkumar Ghodake:** Design of a Mechanical Filter for Nonlinear Ultrasonic Measurements – Published conference paper from the 2nd National Conference on Multidisciplinary Design, Analysis, and Optimization (MDAO 2019). Co-authored with S. Chavan and S.

Kulkarni, this research details the design of a structural filter composed of a periodic, alternate bonded disk arrangement of differing acoustic impedances. It utilizes one-dimensional wave analysis and finite element simulations to formulate an optimization problem for disk thicknesses, isolating damage-induced material response from instrumentally corrupted input stage harmonics.

Source: https://www.researchgate.net/publication/357015228_Design_of_a_Mechanical_Filter_for_Nonlinear_Ultrasonic_Measurements

- **ASA Virtual Student Summer Talks Topology Optimization – Pravinkumar Ghodake Topology Optimization Framework:** Preliminary Results of Topology Optimization for Design of Mechanical Filter in Nonlinear Ultrasonic Testing – Presentation detailing a topology optimization framework used to design a mechanical filter made of alternating bonded material disks. The design leverages 1D wave analysis to optimize disk thicknesses, systematically isolating pure damage-induced material data from instrument-corrupted input stage harmonics.

  Source: https://www.researchgate.net/publication/354423821_Preliminary_Results_of_Topology_Optimization_for_Design_of_Mechanical_Filter_in_Nonlinear_Ultrasonic_Testing

- **ResearchGate Archive (JASA Inverse Problems Selection) – Pravinkumar Ghodake:** Damage size quantification in one-dimensional bar using nonlinear ultrasonics – Published journal abstract exploring the formulation of inverse optimization problems to isolate localized quadratic and cubic material damage sizes from synthetic and noise-corrupted datasets utilizing gradient and non-gradient optimization methods.

  Source: https://www.researchgate.net/publication/347309837_Damage_size_quantification_in_one-dimensional_bar_using_nonlinear_ultrasonics

- **ResearchGate Archive (JASA Wave Propagation Hysteresis) – Pravinkumar Ghodake & Saurabh Biswas:** Wave propagation in a one-dimensional bar with rate-independent hysteresis – Published journal abstract, co-authored with S. Biswas, detailing a numerical spring-mass chain simulation comparing the scalar Bouc-Wen model against a two-state hysteresis model to map micro-structural fatigue and minor loop partial reversals in an aluminum bar under nonlinear wave propagation.

  Source: https://www.researchgate.net/publication/347465881_Wave_propagation_in_a_one-dimensional_bar_with_rate-independent_hysteresis

- **ResearchGate Archive (JASA Functionally Graded Symmetric Media) – Pravinkumar Ghodake:** Nonlinear bulk wave in a functionally graded symmetric hysteretic material model—A numerical study – Published journal abstract detailing a numerical spring-mass chain framework implemented to simulate nonlinear wave propagation through a functionally graded hysteretic medium, evaluating harmonic generation under single-frequency forward and reflected wave conditions.

  Source: https://www.researchgate.net/publication/352068263_Nonlinear_bulk_wave_in_a_functionally_graded_symmetric_hysteretic_material_model-A_numerical_study

- **DEStech Transactions on Structural Health Monitoring (SHM 2021, Stanford University) – Pravinkumar Ghodake:** This paper presents finite element studies on the harmonic scattering of SH waves from localized material nonlinearities and early-stage damage (such as local plasticity and micro-cracks) in metallic components. While experimental studies are challenging due to low backscattered amplitudes and transducer requirements, the finite element analysis demonstrates characteristic harmonic scattering patterns, verifying backscattered Lamb wave generation and examining one-way and two-way wave mixing effects. The findings indicate that these harmonically scattered wave characteristics can help quantify the intensity, size, and position of localized damage.

  Source: https://dpi-proceedings.com/index.php/shm2021/article/view/36362

- **The Journal of the Acoustical Society of America Abstract (JASA 2021) – Pravinkumar Ghodake:** A simple spring-mass chain is implemented to model nonlinear wave propagation through a functionally graded hysteretic material. Harmonic generation due to single-frequency forward and reflected waves, along with the corresponding hysteretic curves, is systematically discussed to analyze dynamic material changes.

  Source: https://www.researchgate.net/publication/352068263_Nonlinear_bulk_wave_in_a_functionally_graded_symmetric_hysteretic_material_model-A_numerical_study

- **The Journal of the Acoustical Society of America Abstract (JASA 2021) – Pravinkumar Ghodake:** This study models localized early-stage materials damage (such as plastic deformation and slip bands) in metals under fatigue as symmetric and asymmetric hysteretic non-linear inclusions within a 1D spring-mass chain configuration. The results reveal that symmetric models generate exclusively odd harmonics in forward and backscattered wave fields, whereas asymmetric models produce both odd and even harmonics. Additionally, energy transitions from fundamental frequencies to higher spectral terms as local damage scales up, while wave field reflections at damage interfaces exhibit non-linear shifting amplitudes caused by wave interference.

  Source: https://www.researchgate.net/publication/352093571_One-dimensional_harmonic_scattering_from_a_localized_damage_modeled_as_symmetric_and_asymmetric_hysteretic_nonlinearity

- **UKACM Conference Proceedings (UKACM 2021) – Pravinkumar Ghodake:** This computational study examines wave mixing in a 1D spring-mass chain system incorporating Reid's hysteresis model to evaluate early-stage micro-structural fatigue and local plasticity. By transmitting two consecutive Gaussian pulses with a specific time delay and amplitude ratio, the research leverages the material's memory effect to generate unique sum and difference frequencies (($3f_1 + f_2$) and ($f1 + 3f_2$)). This computationally efficient approach isolates selective mixed frequencies, avoiding the broad spectrum typically generated by single-pulse mixed-frequency methods.

  Source: https://www.researchgate.net/publication/352226528_Wave-mixing_due_to_two_consecutive_short_pulses_in_Reid's_pinched_hysteretic_material

- **International Congress on Sound and Vibration Proceedings (ICSV 2021) – Pravinkumar Ghodake:** This conference paper investigates one-dimensional wave propagation through media featuring asymmetric, rate-independent hysteretic elements, simulated using a discretized spring-mass chain. While traditional symmetric hysteresis models generate exclusively odd harmonics, this study implements an asymmetric Bouc-Wen model to demonstrate that the interaction of a single-frequency ultrasonic wave with asymmetric hysteretic non-linearities generates both odd and even harmonics, reflecting the structural tracking of specialized dislocation substructures and micro-cracks in fatigued metals.

  Source:
  https://www.researchgate.net/publication/354423047_Wave_propagation_in_a_media_with_asymmetric_rate-independent_hysteretic_nonlinearity

- **16th U.S. National Congress on Computational Mechanics Proceedings (USNCCM 2021) – Pravinkumar Ghodake:** This research paper presents a numerical study exploring the interaction of single- and dual-frequency sinusoidal and Gaussian input pulses with a pinched hysteretic material model, discretized as a long chain of spring-mass elements with a parallel hysteretic spring. By implementing Reid's rate-independent pinched model (1956), the framework efficiently captures pinched and non-pinched loop dynamics stemming from micro-crack updates and internal friction in metals and rocks while circumventing numerical singularities.

  Source: https://www.researchgate.net/publication/356264617_One_Dimensional_Nonlinear_Elastic_Wave_Propagation_in_a_Reid's_Rate-independent_Pinched_Hysteretic_Material

- **48th Annual Review of Progress in Quantitative Nondestructive Evaluation Proceedings (QNDE 2021) – Pravinkumar Ghodake:** This conference paper explores one-dimensional nonlinear wave propagation through a symmetric pinched rate-independent hysteretic material, simulated using a spring-mass chain integrated with a parallel hysteresis model proposed by Biswas (2016). The computational experiments demonstrate that single-frequency waves generate only odd harmonics within evolving

symmetric pinched loops. Furthermore, one-way two-wave mixing configurations reveal sum and difference frequencies along with small, symmetric pinched tracking loops rather than the traditional minor loop paths seen in standard Bouc-Wen or two-state models.

Source: https://www.researchgate.net/publication/356264307_One_Dimensional_Nonlinear_Wave_Propagation_in_a_Rate_Independent_Pinched_Hysteretic_Material

- **25th International Congress of Theoretical and Applied Mechanics Proceedings (ICTAM 2021) – Pravinkumar Ghodake & Saurabh Biswas:** This co-authored research paper presents a systematic evaluation of one-dimensional, nonlinear one-way two-wave mixing across rate-independent hysteretic mediums utilizing discretized spring-mass chains. By implementing both the classical scalar Bouc-Wen model and a newer, multi-state hysteresis model, the investigation accurately isolates sum and difference mixed harmonics alongside fundamental odd spectral components under dual-frequency Gaussian pulse scenarios, offering deeper insights into multi-frequency non-destructive material damage characterization.

  Source: https://www.researchgate.net/publication/354423054_Nonlinear_one-way_two-wave_mixing_in_a_media_with_rate_independent_hysteresis

- **25th International Congress of Theoretical and Applied Mechanics Poster Repository – Pravinkumar Ghodake & Saurabh Biswas:** This technical conference poster outlines additional dataset visuals and structural wave trends from a 1D, nonlinear one-way two-wave mixing analysis. Incorporating both the classic scalar Bouc-Wen model and a newer two-state friction-based hysteresis formulation, the poster illustrates the precise capture of minor, intermediate, and major hysteretic loading-unloading tracking loops under dynamic dual-frequency Gaussian pulse excitations.

  Source: https://www.researchgate.net/publication/361282474_Nonlinear_One-Way_Two-Wave_Mixing_in_a_Media_with_Rate_Independent_Hysteresis_Additional_Data_as_Poster

- **ASME International Design Engineering Technical Conferences Proceedings (IDETC/CIE 2021) – Pravinkumar Ghodake:** This research paper presents a numerical wave propagation study using a 1D spring-mass chain model to capture harmonic scattering from two distinct nonlinear contact interfaces separated by a linear material. By modeling the contact joints using symmetric (Bouc-Wen) and asymmetric (Generalized Bouc-Wen) rate-independent hysteresis frameworks, the analysis demonstrates that symmetric interfaces yield strictly odd harmonics in backscattered and forward wave fields, whereas asymmetric interfaces generate mixed odd and even harmonic spectral distributions.

  Source: https://www.researchgate.net/publication/356264396_Harmonic_Scattering_Of_Elastic_Wave_From_Nonlinear_Contact_Interfaces_Separated_by_a_Linear_Material

- **IEEE International Ultrasonics Symposium Proceedings (IUS 2021) – Pravinkumar Ghodake:** This research paper presents a numerical study investigating contact acoustic nonlinearity (CAN) caused by microscale events like crack clapping and sliding at macroscopic scales. Utilizing a 1D long chain of spring masses, the study evaluates localized layered damages modeled via symmetric (Bouc-Wen) and asymmetric (Generalized Bouc-Wen) frameworks. The computational results demonstrate that scaling up the stiffness parameters of the asymmetric hysteretic layers directly increases even-harmonic amplitudes across both backscattered and forward scattered bulk wave fields, allowing precise evaluation of stacked contact defects.

  Source: https://www.researchgate.net/publication/356264334_Harmonic_Scattering_of_an_Ultrasonic_Wave_from_a_Localized_Layered_Hysteretic_Damage

- **The Journal of the Acoustical Society of America Abstract (JASA 2021) – Pravinkumar Ghodake:** This publication addresses the highly non-linear nature of forward problems in a 1D bar by introducing uncertainty quantification using Reliability-Based Design Optimization (RBDO) and Bayesian Inverse (BI) methods. To determine localized quadratically non-linear damage sizes accurately despite parameters variation (such as input frequency and linear wave velocity), the research integrates Quantile Monte Carlo (QMC) approaches alongside surrogate models like Support Vector Regression

(SVR), Polynomial Chaos Expansions (PCE), and Kriging to drastically minimize total computational execution times.

Source: https://www.researchgate.net/publication/356364243_Uncertainty_quantification_of_quadratically_nonlinear_local_damage_size_quantification_using_nonlinear_ultrasonics

- **The Journal of the Acoustical Society of America Abstract (JASA 2021) – Pravinkumar Ghodake:** This publication details numerical finite element method (FEM) simulations tracking the harmonic scattering of A_0 Lamb waves from localized, early-stage metallic degradation modeled using the Murnaghan hyperelastic formulation for aluminum. The computational studies successfully capture forward and backscattered displacements at the plate surface, mapping out-of-phase reflections alongside complex sum, difference, and high-frequency mode conversions arising during one-way and two-way two-wave structural guided mixing sweeps.

  Source: https://www.researchgate.net/publication/356367329_Discussion_on_harmonic_scattering_of_A0_wave_from_a_local_nonlinear_damage-A_numerical_study

- **The Journal of the Acoustical Society of America Abstract (JASA 2021) – Pravinkumar Ghodake:** This research introduces a systematic size optimization framework using the Nelder-Mead algorithm and COMSOL finite element simulations to design phononic metamaterials that filter out instrument-induced, system-generated higher harmonics. By accounting for realistic boundary conditions and material geometric nonlinearities, the optimized layered configurations successfully isolate pure material damage responses under single-frequency waves and one-way two-wave mixing, suppressing unwanted second harmonic ($2f$) noise amplitudes by up to 50 to 80 times.

  Source: https://www.researchgate.net/publication/356372510_Effective_design_of_metamaterial_to_stop_system_generated_harmonics_during_nonlinear_ultrasonic_testings

- **4th National Conference on Multidisciplinary Analysis and Optimization Proceedings (NCMDAO 2021, IIT Madras) – Pravinkumar Ghodake:** This publication outlines an inverse design framework established to suppress parasitic, instrument-induced second harmonics (2*f*) that typically obscure early-stage material damage evaluations during non-destructive evaluations. Moving past classical trial-and-error approaches, the methodology treats phononic crystal lattice tuning as a structured size optimization problem, simulating geometric nonlinearity and realistic immersion boundary conditions to engineer highly effective structural bandgaps.

  Source: https://www.researchgate.net/publication/360909747_Inverse_Design_to_Stop_Unwanted_Second_Harmonics_in_Nonlinear_Ultrasonics

- **4th National Conference on Multidisciplinary Analysis and Optimization Proceedings (NCMDAO 2021, IIT Madras) – Pravinkumar Ghodake:** This conference publication presents an optimization framework aimed at quantifying the physical size of localized, early-stage material defects (such as dislocation networks and micro-voids) using non-linear guided wave mixing parameters. The inverse design methodology maps non-linear harmonic interaction fields to isolate damage dimensions from structural geometry constraints, minimizing computational iteration demands through automated forward-problem sweeps.

  Source: https://www.researchgate.net/publication/356264050_Early-Stage_Local_Damage_Size_Quantification_using_Nonlinear_Wave_Mixing

- **XLII CILAMCE / III PANACM Conference Proceedings (Rio de Janeiro, Brazil, 2021) – Pravinkumar Ghodake:** This conference paper provides a numerical perspective on the reflection of nonlinear elastic waves in media containing symmetric pinched hysteretic elements modeled using Reid's constitutive relations. Discretized as a 1D spring-mass chain, the simulation analyzes wave interactions at both free and fixed boundaries. The study reveals that the reflected nonlinear wave transfers energy from higher odd harmonics (5*f* and 7*f*) down to the 3rd harmonic (3*f*), while boundary reflections explicitly open the pinch point at the origin of the evolving hysteretic loops.

Source: https://www.researchgate.net/publication/356264251_Reflection_of_Nonlinear_Waves_in_Reid's_Hysteretic_Material_A_Numerical_Perspective

- **66th Congress of the Indian Society of Theoretical and Applied Mechanics Proceedings (ISTAM 2021) – Pravinkumar Ghodake:** This computational paper demonstrates the harmonic scattering of S0 Lamb waves from localized, early-stage structural damages modeled as non-linear elastic inclusions within thin plates. Employing numerical simulations for single-frequency interactions and dual-frequency one-way two-wave mixing configurations, the study reveals that backscattered and forward-scattered harmonic wave fields are highly sensitive to input frequencies, micro-void parameters, and defect sizes, drawing a definitive behavioral analogy to optical 'Raman Scattering' physics.

  Source: https://www.researchgate.net/publication/356786945_Harmonic_Scattering_of_S0_Lamb_Wave_-_A_Computational_Study

- **66th Congress of the Indian Society of Theoretical and Applied Mechanics Proceedings (ISTAM 2021) – Pravinkumar Ghodake:** This open-access publication derives analytical theoretical solutions to demonstrate the critical sensitivity of wave fields to local stiffness reductions caused by micro-void accumulation and dislocation breakaways. By modeling the defect as a localized quadratically non-linear material, the investigation details the exact continuity conditions, forward-harmonic generation mechanisms, and compensatory wave interactions that dictate non-intuitive energy transfers between fundamental (1st) and second (2nd) harmonics across localized degradation zones spanning 0 to 20 mm.

  Source: https://www.researchgate.net/publication/356916787_Effect_of_Loss_in_Local_Stiffness_on_Harmonic_Scattering_of_Longitudinal_Wave_from_a_Quadratically_Nonlinear_Local_Damage

- **ISNDT Annual Conference Proceedings (NDE 2021) – Pravinkumar Ghodake:** This conference paper targets the elimination of parasitic, instrument-induced higher harmonics that mask pure early-stage material damage responses. It structures a rigorous shape

optimization problem using non-gradient algorithms to design steel-and-glass layered phononic metamaterials. By automating an inverse coupling loop between MATLAB and COMSOL Multiphysics, the optimized geometric configurations establish robust bandgaps that suppress 2nd (2*f*), 3rd (3*f*), and 4th (4*f*) system harmonics while ensuring energy does not leak into fractional tracking intervals.

Source: https://www.researchgate.net/publication/356914386_Shape_Optimization_for_the_Design_of_Linear_and_Nonlinear_Metamaterials_to_Improve_Nonlinear_Ultrasonic_Testing

- **ISNDT Annual Conference Proceedings (NDE 2021) – Pravinkumar Ghodake:** This computational work implements a highly efficient 1D spring-mass chain model composed of 2,000 discrete elements to track structural acoustic interactions with localized, non-linear defects. By modeling crack clapping, fatigue, and interface sliding mechanisms through Reid's pinched hysteresis formulation in MATLAB—while strictly maintaining convergence via the CFL condition—the study maps how expanding damage lengths (ranging across 20 to 120 masses) actively modulates backscattered and forward scattered wave field signatures.

  Source: https://www.researchgate.net/publication/356960740_Computational_Study_on_Interaction_of_Nonlinear_Waves_with_a_Localized_Reid's_Pinched_Hysteretic_Damage

- **ASME International Design Engineering Technical Conferences Proceedings (IDETC/CIE 2021) – Pravinkumar Ghodake:** This research paper presents a numerical wave propagation study using a 1D spring-mass chain model to capture harmonic scattering from two distinct nonlinear contact interfaces separated by a linear material. By modeling the contact joints using symmetric (Bouc-Wen) and asymmetric (Generalized Bouc-Wen) rate-independent hysteresis frameworks, the analysis demonstrates that symmetric interfaces yield strictly odd harmonics in backscattered and forward wave fields, whereas asymmetric interfaces generate mixed odd and even harmonic spectral distributions.

  Source: https://www.researchgate.net/publication/356264396_Harmonic_Scattering_Of_Elastic_Wave_From_Nonlinear_Contact_Interfaces_Separated_by_a_Linear_Material

- **The Journal of the Acoustical Society of America Abstract (JASA 2022) – Pravinkumar Ghodake:** This publication details the inverse design of linear periodic metamaterials capable of simultaneously controlling both longitudinal and transverse waves, including their second harmonics, under monochromatic wave propagation and one-way two-wave mixing conditions. By framing the lattice design parameters—including layer widths and the integer number of repeated periodic cells (N)—within a constrained, gradient-free optimization algorithm and utilizing finite element analysis, the framework successfully solves the multi-wave inverse problem within a few minutes under strong structural constraints.

  Source: https://www.researchgate.net/publication/360481440_Metamaterials_to_control_both_longitudinal_and_transverse_waves_simultaneously

- **The Journal of the Acoustical Society of America Abstract (JASA 2022) – Pravinkumar Ghodake:** This publication introduces a time-dependent shape optimization framework using a non-gradient algorithm and finite element modeling to design cylindrical metamaterials for wave control in cylindrical rods. By targeting single and multi-objective functions to suppress axial and radial displacements across end surfaces from monochromatic Gaussian pulses, the methodology determines optimal widths for layered elastic materials arranged periodically along subsurfaces and outer surfaces. The model integrates weak total-length constraints to ensure that every computationally generated design iteration remains readily manufacturable without requiring prior bandgap mapping.

  Source: https://www.researchgate.net/publication/360489177_Design_optimization_of_metamaterials_to_control_waves_in_cylindrical_rods

- **10th International Conference on Inverse Problems: Modeling and Simulation Proceedings (IPMS 2022, Malta) – Pravinkumar Ghodake:** This conference paper presents an inverse optimization formulation designed to estimate the size of a localized quadratically non-linear damage zone sandwiched between two linear elastic materials. Because classical single-frequency higher-harmonic generation techniques possess narrow sizing limits, this research utilizes a 1D non-linear elastic one-way two-wave mixing

configuration to derive approximate analytical solutions for both backscattered and transmitted fields, establishing a robust inverse problem framework that successfully broadens defect characterization bounds.

Source: https://www.researchgate.net/publication/363257701_Inverse_problem_for_quantification_of_localized_damage_using_1D_nonlinear_elastic_two-wave_mixing

- **22nd International Symposium on Nonlinear Acoustics Proceedings (ISNA 2022, University of Oxford) – Pravinkumar Ghodake:** This conference paper evaluates wave field interactions across periodically distributed non-linear inclusions (circular and square shapes) utilizing advanced finite element models. The computational studies reveal a notable self-cancellation phenomenon where carefully optimized inclusion dimensions and intermediate lattice spacing configurations completely eliminate the transmission of 2nd higher harmonics due to wave interference and local energy trapping, simulating an "elastically invisible" non-linear metamaterial network.

  Source: https://www.researchgate.net/publication/363257744_Do_Periodic_arrangement_of_Nonlinear_Local_Damages_act_as_Nonlinear_Metamaterials

- **12th International Conference on Metamaterials, Photonic Crystals and Plasmonics Proceedings (META 2022, Torremolinos, Spain) – Pravinkumar Ghodake:** This publication details the inverse design of a unique direction-dependent mechanical metamaterial engineered to simultaneously and independently control longitudinal waves moving in one direction and transverse waves propagating in the exact opposite direction. By formulating time-dependent inverse optimization problems solved via gradient-free algorithms, the methodology minimizes parasitic second harmonics ($2f$ = 4MHz) while maximizing fundamental harmonic transmission (f = 2MHz) to maintain peak power within short Gaussian pulse tracking windows.

  Source: https://www.researchgate.net/publication/363257827_Inverse_Design_of_Direction_Dependent_Mechanical_Metamaterial

- **13th International Workshop on Structural Health Monitoring Proceedings (IWSHM 2021/2022, Stanford University) – Pravinkumar Ghodake:** This publication details advanced finite element simulations tracking the harmonic scattering of SH waves from localized, micro-scale damage zones such as plastic deformation zones and shear bands in metallic plates. The numerical model maps specific acoustic footprints, verifying that forward-scattered SH waves exhibit exclusively odd harmonics, while both forward and backscattered mode-converted Lamb waves reveal distinct even harmonic spectral signatures and heightened static field components.

  Source: https://www.researchgate.net/publication/363484628_Harmonic_Scattering_of_SH_Waves_from_a_Localized_Damage_Finite_Element_Studies

- **13th International Workshop on Structural Health Monitoring Proceedings (IWSHM 2021/2022, Stanford University) – Pravinkumar Ghodake:** This conference paper presents a numerical study investigating the interaction of single-frequency and two-frequency (one-way two-wave mixing) ultrasonic waves with randomly distributed, localized hysteretic damages. Utilizing a 1D spring-mass chain configuration, the research implements the classic Bouc-Wen model for symmetric hysteretic elements and the Generalized Bouc-Wen model for asymmetric hysteretic elements. The computational results demonstrate that randomly distributed symmetric damage fields yield exclusively odd harmonics, whereas asymmetric tracking distributions generate a mixed spectrum of both odd and even higher harmonic parameters.

  Source: https://www.researchgate.net/publication/363484751_Nonlinear_Bulk_Wave_Propagation_in_a_Material_with_Randomly_Distributed_Symmetric_and_Asymmetric_Hysteretic_Nonlinearity

- **IMPLAST 2022 Conference Proceedings (IIT Madras) – Pravinkumar Ghodake:** This conference paper evaluates the interactions of non-linear ultrasonic waves within discontinuous, functionally graded non-linear materials using finite element analysis (FEM). Moving past idealized uniform damage assumptions, the computational experiments evaluate monochromatic wave propagation (f = 4 MHz) alongside dual-frequency one-way two-wave mixing sequences under both free and fixed boundary

conditions, mapping how spatial degradation at the microscale triggers complex concurrent harmonic generation and scattering mechanisms.

Source: https://www.researchgate.net/publication/380348060_Nonlinear_Waves_in_Discontinuous_Functionally_Graded_Nonlinear_Materials

- **IMPLAST 2022 Conference Proceedings (IIT Madras) – Pravinkumar Ghodake:** This publication introduces a time-domain inverse design strategy that shifts away from standard analytical transfer matrix infinite-length assumptions and eigenvalue sweeps. By proposing time-dependent objective functions integrated with non-gradient optimization routines and finite element modeling, the framework successfully controls both the time and frequency responses of axial displacements (u_z) through a cylindrical rod, circumventing multi-modal cross-coupling between radial and axial wave metrics.

  Source: https://www.researchgate.net/publication/380348225_Inverse_Design_of_Linear_Mechanical_Metamaterial_to_Control_Wave_Propagation_in_Elastic_Rod

- **Inter-Noise 2022 Conference Proceedings (Glasgow, UK) – Pravinkumar Ghodake:** This publication leverages transient finite element analysis paired with gradient-free optimization algorithms—such as coordinate search and the Nelder-Mead method—to engineer finite-length linear and nonlinear cylindrical metamaterial rods. By transitioning away from infinite-length idealizations, the modeling framework accounts for material and geometric nonlinearities to mitigate harmonic scattering and isolate pure damage signatures, successfully filtering out instrument-induced 2^{nd} harmonics while maximizing fundamental short-pulse propagation in longitudinal cylindrical modes.

  Source: https://www.researchgate.net/publication/363519765_Inverse_Design_of_Linear_and_Nonlinear_Cylindrical_Metamaterial_Rod

- **International Conference on Imaging NDE Proceedings (ICINDE 2022) – Pravinkumar Ghodake:** This conference publication explores the computational complexity of harmonically scattered waves from clusters of localized defects, modeled as multiple solid and elastically hollow circular non-linear inclusions. By carrying out non-

linear finite element method (FEM) simulations, the study maps complex multi-modal interactions involving concurrent harmonic generation, mode conversion, and linear interference near boundaries, capturing the unique presence and absence of fractional-order higher harmonics (1.5*f*, 2.5*f*, and 3.5*f*).

Source: https://www.researchgate.net/publication/365295228_Presence_and_Absence_of_Fractional_Harmonics_in_Scattered_Waves_from_Multiple_Nonlinear_Inclusions

- **The Journal of the Acoustical Society of America Abstract (JASA 2022) – Pravinkumar Ghodake:** This publication details numerical finite element studies investigating the reflection of nonlinear ultrasonic waves within continuous functionally graded materials to model non-destructive pulse-echo testing of pressure vessels. The simulation platform updates linear and nonlinear constitutive equations to map linear and exponential spatial variations of material parameters, density, and local stiffness losses. The results show that reflected longitudinal higher harmonics (2*f*, 3*f*, 4*f*, ...) exhibit increased energy profiles from fixed boundaries compared to free boundaries and are highly sensitive to spatial parameter tracking layouts.

  Source: https://www.researchgate.net/publication/365844989_Finite_element_studies_to_understand_reflection_of_nonlinear_waves_in_a_continuous_functionally_graded_material

- **The Journal of the Acoustical Society of America Abstract (JASA 2022) – Pravinkumar Ghodake:** This publication details an inverse design strategy formulated to engineer direction-independent elastic metamaterials that can independently control longitudinal and transverse waves for non-destructive testing applications. By setting up time-dependent, min-max multi-objective optimization problems solved via gradient-free algorithms, the forward transient finite element models systematically attenuate parasitic, instrument-induced second harmonics (2*f*) while maximizing fundamental output pulse power (*f*).

  Source: https://www.researchgate.net/publication/365845648_Inverse_design_of_direction_independent_metamaterials_to_control_longitudinal_and_transverse_waves

- **INTER-NOISE and NOISE-CON Congress and Conference Proceedings – Pravinkumar Ghodake:** This publication applies transient finite element shape optimization to engineer linear and nonlinear cylindrical metamaterial rods of finite lengths. Rather than relying on infinite-length assumptions, the modeling framework integrates both material and geometric nonlinearities to account for harmonic scattering. Utilizing gradient-free algorithms like coordinate search and Nelder-Mead, the design successfully filters parasitic 2^{nd} harmonics while optimizing the transmission of fundamental short-pulse longitudinal modes.

  Source: https://www.researchgate.net/publication/368536898_Inverse_Design_of_Linear_and_Nonlinear_Cylindrical_Metamaterial_Rod

- **The Journal of the Acoustical Society of America Abstract (JASA 2023) – Pravinkumar Ghodake:** This publication introduces a novel architectural unit cell composed of three adjacent non-linear elastic layers designed to function as a tunable mechanical metamaterial. Building on classical elastodynamic reciprocity and compensatory wave principles, the finite element modeling framework adjusts the physical width parameters and central layer non-linear property intensities to precisely modulate, trap, or eliminate the transmission of 2^{nd} higher harmonics ($2f$) across structural forward and backscattered wave fields.

  Source: https://www.researchgate.net/publication/370358539_Unit_cell_of_three_nonlinear_layers_for_the_design_of_tunable_nonlinear_metamaterials

- **Invisible Nonlinear Inclusions – APS March Meeting Proceedings (Superlattices and Artificially Structured Materials) – Pravinkumar Ghodake:** This critical conference publication outlines a time-dependent constrained optimization framework developed to achieve true **nonlinear cloaking** and **invisibility** against longitudinal ultrasonic wave fields. By modeling complex, localized clusters of elliptical hyperelastic inclusions (Murnaghan material) embedded in a linear matrix, the design variables controlling inclusion size, shape, rotation angle, and relative spatial position are tuned via the Nelder-Mead gradient-free algorithm. The resulting optimal structural cluster demonstrates total

self-cancellation of transmitted higher harmonics, rendering complex fatigue-damaged regions elastically invisible to incoming longitudinal diagnostic waves.

Source: https://www.researchgate.net/publication/368881192_Nonlinear_Inclusions_Invisible_to_Longitudinal_Ultrasonic_Wave

- **APS March Meeting Poster Session on Additive Manufacturing NDE – Pravinkumar Ghodake:** This conference publication presents a finite element method (FEM) framework investigating structural health monitoring in single- and multi-material additively manufactured metals. To address non-destructive evaluation challenges introduced by layered depositions and microscale interface defects (such as micro-voids and local plasticity), the study carries out numerical experiments analyzing monochromatic wave paths alongside one-way and two-way two-wave mixing configurations. The results reveal that the dynamic interplay between linear and nonlinear impedance mismatches produces unique forward and backscattered harmonic wave responses.

  Source: https://www.researchgate.net/publication/368881181_Harmonic_Scattering_of_Nonlinear_Waves_from_Layered_Materials_that_mimics_Additively_Manufactured_Metals

- **NODYCON Conference Presentation on Crossed-Thin-Rectangular Non-Linear Inclusions – Pravinkumar Ghodake:** This computational study leverages the finite element method to circumvent limitations in theoretical techniques regarding complex wave fields. By analyzing unique, custom-shaped crossed-thin-rectangular inclusions, the research maps parameter controls over inclusion geometry and spatial arrays. The resulting models demonstrate explicit non-linear wave manipulation, illustrating energy exchange and mode conversion trends between harmonically scattered longitudinal and transverse waves to aid in advanced mechanical metamaterial design.

  Source: https://www.researchgate.net/publication/371691997_Harmonic_Scattering_of_Waves_from_Crossed-thin-rectangular_Nonlinear_Inclusions

- **NODYCON Conference Paper on Inverse Metamaterial Optimization – Pravinkumar Ghodake:** This publication details the configuration of periodic and quasi-periodic mechanical metamaterials designed to regulate non-linear elastic wave fronts. To address multi-parameter complexities introduced by concurrent harmonic generation and scattering fields, the workflow frames the layer width mapping as a time-dependent optimization problem. The forward transient problem is analyzed using finite element modeling, and the inverse geometry optimization is resolved using the Nelder-Mead algorithm to suppress unwanted diagnostic harmonics.

  Source:
  https://www.researchgate.net/publication/371984410_Inverse_Design_of_Periodic_and_Quasi-Periodic_Nonlinear_Mechanical_Metamaterial

- **META Conference Paper on Time-Varying Elastic Metamaterials – Pravinkumar Ghodake:** This publication documents numerical experiments designed to investigate the emerging non-linear effects produced by linear, time-dependent elastic metamaterials. By modulating structural material parameters dynamically across the time domain, the research details advanced wave manipulation capabilities. These include the automated generation of higher harmonics, precise control over scattered spectral components, and the regulation of mixed-frequency combinations during single-frequency and two-wave guided mixing sequences.

  Source:
  https://www.researchgate.net/publication/372317065_Nonlinear_Effects_of_Linear_Time-Dependent_Metamaterials

- **META Conference Paper on Functionally Graded Metamaterial Unit Cells – Pravinkumar Ghodake:** This research paper presents a series of computational experiments analyzing a novel non-linear, functionally graded metamaterial unit cell embedded within linear elastic media of matching impedance. By evaluating the sensitivities of spatial parameter configurations under forward and inverse design approaches, the modeling framework maps how the continuous distribution of non-linear properties manipulates and structures the harmonic responses of propagating elastic waves.

Source: https://www.researchgate.net/publication/372317060_Introduction_to_Functionally_Graded_Unit_Cell_of_Nonlinear_Metamaterial_that_Controls_Harmonic_Responses_of_Elastic_Waves

- **Time Crystals – IEEE International Ultrasonics Symposium Poster on Time-Varying Phononic Crystals – Pravinkumar Ghodake:** Drawing strategic parallels between 4D optics and acoustic wave mechanics, this research poster explores non-linear elastic wave propagation inside time-varying linear and non-linear phononic metamaterials. By modulating the material's linear and non-linear physical properties dynamically in time, the underlying finite element model captures complex wave scattering responses, including the manipulation of non-integer fractional harmonics. The study establishes the capacity of transient inverse design frameworks to regulate scattered wave fields across time-dependent media.

  Source: https://www.researchgate.net/publication/375407198_Ultrasonic_Waves_in_Time-Varying_Nonlinear_Phononic_Crystals

- **The Journal of the Acoustical Society of America Abstract on Functionally Graded Inclusions – Pravinkumar Ghodake:** This publication details numerical experiments mapping wave actions inside localized functionally graded non-linear inclusions to outline building blocks for tunable metamaterials. The computational results demonstrate a unique invariance property where distinct spatial distributions of non-linear parameters produce identical forward and backscattered harmonic responses during monochromatic wave interactions and one-way two-wave mixing, provided the integrated area beneath the spatial distribution curve remains constant.

  Source: https://www.researchgate.net/publication/376018222_Interesting_effects_of_harmonically_scattered_waves_from_1D_functionally_graded_nonlinear_inclusions

- **The Journal of the Acoustical Society of America Abstract on 2-D Corner Geometric Scattering – Pravinkumar Ghodake:** This publication examines the intricate wave mechanics of reflected non-linear ultrasonic fields interacting with inaccessible surfaces, modeling triangular, circular, and rectangular 2-D corners via the finite element method. The extensive numerical experiments map a complex elastodynamic interplay where boundary boundaries trigger concurrent harmonic generation, linear scattering, mode conversion between bulk longitudinal and transverse wave components, and spectral energy redistribution during monochromatic and two-wave mixing sweeps.

  Source: https://www.researchgate.net/publication/376020701_The_complexity_of_harmonically_scattered_nonlinear_waves_from_triangular_circular_and_rectangular_corners_of_the_2-D_domain

- **ISNDT Annual Conference Proceedings on Asymmetric Functionally Graded Inclusions (NDE 2023, Pune) – Pravinkumar Ghodake:** This publication details numerical analyses of 1D asymmetric, functionally graded non-linear inclusions ranging from 2 to 10 mm in size to simulate microstructural defects in metals. Moving away from traditional uniform damage assumptions, the computational model features non-linear parameters that follow an asymmetric profile along the spatial coordinate. The study maps how these unique gradient tracking layouts alter harmonic scattering and generate a distinct spectrum of higher harmonics alongside localized static component variations.

  Source: https://www.researchgate.net/publication/380348069_Numerical_Analysis_of_Asymmetrical_Functionally_Graded_Small-Scale_1D_Inclusions_for_Harmonic_Scattering

- **Gordon Research Seminar (GRS) Presentation on Multi-Functional Multi-Material Metamaterials (Ventura, California) – Pravinkumar Ghodake:** This highly impactful international conference paper documents an advanced inverse design framework engineered to solve multi-objective optimization equations for deep wave control. Moving past standard linear limitations, this research introduces an automated level-set shape optimization method that simultaneously dictates layer width geometries and precise material stacking sequences for 2-to-4 layer configurations. By implementing gradient-free

computational algorithms to map transient finite element models, this breakthrough approach enables the synthesis of highly tunable, directional, and multi-functional mechanical metamaterials tailored to isolate, pass, or block high-frequency (MHz) diagnostic elastic wave fields.

Source: https://www.researchgate.net/publication/378145055_Manipulating_Elastic_Waves_A_Novel_Approach_via_Inverse_Design_of_Multi-Material_Multi-Functional_Metamaterials

- **Gordon Research Conference (GRC) Technical Poster on Multi-Material and Multi-Functional Metamaterials (Ventura, California) – Pravinkumar Ghodake:** This prominently positioned open-science research portfolio documents the implementation of a comprehensive inverse design optimization platform engineered to sculpt both linear and non-linear high-frequency (MHz) elastic waves. The framework incorporates finite lengths, authentic boundary conditions, and level-set shape optimization variables to bypass traditional infinite-periodic restrictions. By running gradient-free mathematical search loops across multi-objective functions, this methodology concurrently tunes spatial physical dimensions and material stacking configurations to synthesize next-generation, direction-dependent, and multi-functional mechanical metamaterials with highly targeted stop and pass bands.

  Source: https://www.researchgate.net/publication/378145129_Elastic_wave_manipulation_through_inverse_design_of_multi-material_and_multi-functional_metamaterials

- **APS March Meeting Technical Poster on Time-Modulated Linear Elastic Unit Cells – Pravinkumar Ghodake:** This strategically indexed conference publication chronicles an investigation into the emerging non-linear response signatures of time-varying phononic media for structural health monitoring applications. The study evaluates a novel architectural unit cell composed of a time-independent linear material sandwiched between two time-modulated linear layers matched in impedance. Using non-linear transient finite element analysis, the research details how adjusting the modulation frequencies shifts the

distribution, presence, and suppression of high-frequency harmonic energy fields via dynamic wave scattering interfaces.

Source:
https://www.researchgate.net/publication/376835312_Harmonic_Mastery_in_Time-Modulated_Linear_Elastic_Unit_Cells

- **APS March Meeting Oral Presentation on SH Wave Harmonic Scattering – Pravinkumar Ghodake:** Supported by the Acoustical Society of America (ASA) CIRE international student grant, this critical conference publication investigates higher harmonic wave signatures generated by multiple localized, microscale damage arrays (dislocations, micro-cracks). By setting up complex transient finite element studies—encompassing monochromatic wave paths along with one-way and two-way two-wave mixing sweeps—the analysis uncovers a unique acoustic response where odd harmonics are completely absent in back and forward scattered fields, while even harmonics and static terms dominate. The research details structural harmonic energy trapping and metamaterial-like behaviors across localized clusters, distinguishing a single defect from split defect profiles to aid in advanced wave manipulation layouts.

  Source: https://www.researchgate.net/publication/376835480_Unraveling_SH_Wave_Harmonic_Scattering_in_Layered_Nonlinear_Localized_Damages

- **The Journal of the Acoustical Society of America Abstract on Symmetric Functionally Graded Unit Cells – Pravinkumar Ghodake:** This publication details the engineering of a functionally graded nonlinear unit cell featuring a spatially continuous, symmetrical distribution of nonlinear parameters across its flanking layers. Building on high-frequency (MHz) acoustic wave interactions and analogies to optical 'Raman Scattering' physics, the research demonstrates how tuning spatial distribution geometries, central layer intensities, and overall cell widths enables non-intuitive, precise control over forward and backscattered wave mechanics—causing specific target harmonics to emerge or vanish completely due to complex internal wave fields.

Source: https://www.researchgate.net/publication/387666539_Exploring_interesting_effects_of_symmetric_functionally_graded_unit_cells_for_the_design_of_nonlinear_mechanical_metamaterials

- **ISNDT Annual Conference Proceedings on Wave Scattering Dynamics in Clusters of Nonlinear Inclusions (NDE 2024, Chennai) – Pravinkumar Ghodake:** Inspired by micro-nano bubbles used in high-resolution contrast imaging, this computational study details a forward design approach to explore harmonic scattering across configurations of solid and elastically hollow non-linear inclusions. The investigation maps out **nonlinear cloaking** traits using strategically positioned solid inclusions to completely eliminate higher harmonics and static terms, contrasting them with elastically hollow nested inclusions that dynamically generate second harmonics ($2f$) and distinct static term configurations at the receiving edge.

  Source: https://www.researchgate.net/publication/387314130_Wave_Scattering_Dynamics_in_Clusters_of_Nonlinear_Solid_and_Elastically_Hollow_Inclusions

- **Nonlinear Wave Cloaking – iNCMDAO International Conference Paper on Inverse Design and Nonlinear Wave Cloaking (IISc Bangalore) – Pravinkumar Ghodake:** This major conference publication maps advanced computational workflows to achieve total **nonlinear wave cloaking** and acoustic invisibility within heterogeneous solid media. To overcome the challenges of modeling complex arrays of incipient defects (such as dislocation networks and local grain plasticity), the framework models multi-scaled elliptical inclusions using a hyperelastic formulation matched in linear impedance. By running gradient-free inverse optimization algorithms on transient finite element models, the setup dynamically coordinates inclusion dimensions, rotation angles, and relative spatial points to suppress parasitic transmitted harmonics, establishing a new design paradigm for **nonlinear cloaking** technologies.

  Source: https://www.researchgate.net/publication/387666302_Inverse_Design_of_Nonlinear_Phononic_Metamaterials_for_Nonlinear_Wave_Cloaking

- **iNCMDAO International Conference Paper on Stacking Optimization for Sculpting Elastic Waves (IISc Bangalore) – Pravinkumar Ghodake:** Presented at the J.N. Tata Auditorium, this high-impact publication establishes a robust time-domain inverse design framework to sculpt high-frequency (MHz) linear and nonlinear elastic waves. Moving beyond conventional eigenvalue sweeps and transfer matrix limits, this methodology implements a level-set shape optimization routine guided by gradient-free algorithms. The platform concurrently optimizes individual layer widths and multi-material stacking sequences for finite-length periodic arrays under realistic boundary conditions, creating highly tunable multifunctional metamaterials capable of directional, mode-independent wave manipulation.

  Source: https://www.researchgate.net/publication/387666456_Sculpting_Linear_and_Nonlinear_Elastic_Waves_Exploring_Inverse_Design_Potentials

- **iNCMDAO International Conference Flyer and Non-Linear Metamaterial Design Summary – Pravinkumar Ghodake:** This open-access technical document presents supplementary materials and reference data associated with the 1st International and 7th National Conference on Multidisciplinary Design, Analysis and Optimization (iNCMDAO). The data highlights underlying computational frameworks engineered for inverse design, exploring the behavior of multi-layered phononic crystal architectures, hyperelastic inclusion arrays, and shape optimization loops used to implement advanced **nonlinear wave cloaking** and acoustic invisibility profiles.

  Source: https://www.researchgate.net/publication/387685334_iNCMDAO_Flyer_UPDATEDpdf

- **iNCMDAO Conference Official Schedule and Technical Program Repository – Pravinkumar Ghodake:** This open-access dataset archive hosts the complete operational schedule and event proceedings for the 1st International and 7th National Conference on Multidisciplinary Design, Analysis and Optimization (iNCMDAO). The programming references specialized tracks detailing shape and topology optimization workflows, hyperelastic material configurations, and computational inverse problem strategies applied

to advance high-frequency elastic wave manipulation and **nonlinear wave cloaking** methodologies.

Source: https://www.researchgate.net/publication/387685229_iNCMDAO_FullProgrammepdf

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Direct Publisher Directory

- **APS Global Physics Summit Oral Presentation on Advanced Metamaterial Design and Acoustic Invisibility – Pravinkumar Ghodake:** Delivered at the APS Global Physics Summit 2025, this oral presentation reveals a major breakthrough in wave control physics through an automated optimization platform tailored for high-frequency elastic media. To overcome the challenges of modeling complex arrays of incipient defects (such as dislocation networks and local grain plasticity), the framework models multi-scaled elliptical inclusions using a hyperelastic formulation matched in linear impedance. By running gradient-free inverse optimization algorithms on transient finite element models, the setup dynamically coordinates inclusion dimensions, rotation angles, and relative spatial points to suppress parasitic transmitted harmonics, establishing a new design paradigm for **nonlinear cloaking** technologies and rendering materials flaws entirely undetectable to diagnostic scans.

  Source: https://meetings-archive.aps.org/smt/2025/mar-t63/9/

- **APS Global Physics Summit Presentation on Time-Dependent Metamaterial Architectures and Harmonic Manipulation – Pravinkumar Ghodake:** Presented at the prestigious APS Global Physics Summit 2025, this research introduces an advanced 4D wave manipulation framework that models time-varying third-order elastic constants within the Murnaghan constitutive framework. By simulating transient finite element interactions between a monochromatic longitudinal bulk wave and a single, time-evolving 2D circular non-linear inclusion, the study uncovers complex local resonance phenomena.

This includes non-intuitive angular energy exchanges between scattered longitudinal and transverse wave modes, alongside the targeted generation of both integer and fractional tracking harmonics to unlock unprecedented spectral design paths.

Source: https://meetings-archive.aps.org/smt/2025/mar-r00/164/

- **The Journal of the Acoustical Society of America (JASA) Article on Symmetric Functionally Graded Unit Cells – Pravinkumar Ghodake:** Published in the *Journal of the Acoustical Society of America*, this research establishes a foundational framework for the architectural design of non-linear mechanical metamaterials. Expanding upon classical elastodynamic models, the study introduces a functionally graded non-linear unit cell characterized by a spatially continuous, symmetrical distribution of material properties. By mathematically mapping and tuning the spatial profiles, cell widths, and central layer property intensities, the simulation platform successfully replicates optical 'Raman Scattering' physics within high-frequency (MHz) acoustic waves. The computational results demonstrate non-intuitive wave mechanics where forward and backscattered harmonics can be engineered to systematically emerge or completely vanish, unlocking powerful new pathways for precise, inverse-designed acoustic filtering and wave control.

  Source: https://pubs.aip.org/asa/jasa/article/156/4_Supplement/A22/3331196/Exploring-interesting-effects-of-symmetric

- **APS March Meeting Oral Presentation on SH Wave Harmonic Scattering and Grant Distinctions – Pravinkumar Ghodake:** Supported by the prestigious Acoustical Society of America (ASA) CIRE international student grant and an institute scholarship from the Indian Institute of Technology Bombay, this oral presentation systematically explores higher harmonic wave actions stemming from multiple localized, microscale damage arrays (dislocations, micro-cracks). By conducting transient finite element case studies across monochromatic wave interactions alongside co-directional and two-way wave mixing sweeps, the research uncovers a unique acoustic response where odd harmonics are entirely suppressed, while even harmonics and static terms dominate the scattered fields. The study maps how geometric clustering induces sub-wavelength harmonic energy

trapping, providing critical insights for structural health monitoring and the engineering of advanced wave-cloaking mechanical metamaterials.

Source: https://meetings-archive.aps.org/mar/2024/qq05/3/

- **NASA ADS Index / APS March Meeting Presentation on Time-Modulated Unit Cells and Harmonic Manipulation – Pravinkumar Ghodake:** Indexed within the prestigious NASA Astrophysics Data System (ADS) and presented at the APS March Meeting 2024, this publication explores emerging non-linear response signatures within time-varying, linear elastic materials. To address the complexities of power-amplified wave interactions in non-linear ultrasonic testing for structural health monitoring, the research evaluates a novel unit cell containing a time-independent core material sandwiched symmetrically between two time-modulated linear boundary layers. Utilizing transient finite element analysis, the computational framework maps how shifting temporal modulation frequencies dictates the selective generation, presence, or absolute suppression of high-frequency harmonic energy fields via dynamic, time-varying scattering interfaces.

  Source: https://ui.adsabs.harvard.edu/abs/2024APS..MARNN00003/abstract

- **The Journal of the Acoustical Society of America (JASA) Article on 1D Functionally Graded Nonlinear Inclusions – Pravinkumar Ghodake:** Published in the journal of the Acoustical Society of America, this research establishes an invariance principle for high-frequency elastic wave mechanics within localized material imperfections. Addressing the inherent limitations of conventional analytical solutions, the study deploys advanced numerical wave experiments to analyze spatial property distributions across 1D inclusions. The findings unveil a crucial physical phenomenon: regardless of the geometric complexity or profile shape of the gradient distribution, forward and backscattered harmonic wave amplitudes remain entirely invariant as long as the mathematical area integrated beneath the spatial non-linear property curve is held constant. This discovery yields a highly predictable structural block for designing tunable non-linear acoustic metamaterials and filtering devices.

Source:
https://pubs.aip.org/asa/jasa/article/154/4_supplement/A58/2923798/Interesting-effects-of-harmonically-scattered

- **The Journal of the Acoustical Society of America (JASA) Article on 2-D Domain Geometric Corner Scattering – Pravinkumar Ghodake:** Published in the *Journal of the Acoustical Society of America*, this publication maps the complex, multi-modal wave mechanics of reflected non-linear ultrasonic fields interacting with inaccessible structural edges. Addressing the critical limitations of classical analytical modeling, the research defines a unique 2-D finite element framework to analyze triangular, circular, and rectangular corner boundaries. The extensive numerical experiments uncover a non-intuitive elastodynamic interplay where boundary boundaries trigger simultaneous harmonic generation, linear scattering, mode conversions between bulk longitudinal and transverse wave components, and comprehensive spectral energy redistribution. These findings are expanded to systematically explore one-way and two-way two-wave mixing configurations, detailing how boundary-induced phase differences isolate fundamental harmonic sensitivities to optimize advanced non-linear pulse-echo testing technologies.

  Source: https://pubs.aip.org/asa/jasa/article/154/4_supplement/A262/2924505/The-complexity-of-harmonically-scattered-nonlinear

- **NASA ADS Index / APS March Meeting Presentation on Additive Manufacturing Non-Destructive Evaluation – Pravinkumar Ghodake:** Indexed within the NASA Astrophysics Data System (ADS) and presented at the APS March Meeting 2023, this publication provides a comprehensive finite element method (FEM) framework investigating structural health monitoring in layered, additively manufactured metallic materials. To address non-destructive evaluation challenges introduced by layered depositions and microscale interface defects (such as micro-voids, micro-cracks, and local plasticity), the study carries out extensive numerical experiments analyzing monochromatic wave paths alongside one-way and two-way two-wave mixing configurations. The research models single- and bi-metallic material configurations in detail, revealing that the dynamic interplay between linear and nonlinear impedance mismatches produces unique forward

and backscattered harmonic wave fields where the backscattered amplitudes can non-intuitively rival forward-scattered fields.

Source: https://ui.adsabs.harvard.edu/abs/2023APS..MARVV1042G/abstract

- **The Journal of the Acoustical Society of America (JASA) Article on Tunable Metamaterial Unit Cells – Pravinkumar Ghodake:** Published in the highly regarded *Journal of the Acoustical Society of America*, this publication introduces a novel architectural unit cell composed of three adjacent non-linear elastic layers designed to function as a tunable mechanical metamaterial. Building on classical elastodynamic reciprocity and compensatory wave principles, the finite element modeling framework adjusts physical width parameters and central layer property intensities. The computational results demonstrate precise modulation over structural forward and backscattered fields, proving that the amplitudes of 2nd higher harmonics ($2f$) can be selectively suppressed or trapped directly at internal material interfaces to achieve advanced non-destructive wave shielding.

  Source: https://pubs.aip.org/asa/jasa/article/153/3_supplement/A166/2885960/Unit-cell-of-three-nonlinear-layers-for-the-design

- **The Journal of the Acoustical Society of America (JASA) Article on Functionally Graded Pulse-Echo Modeling – Pravinkumar Ghodake:** Published in the premier journal of the Acoustical Society of America, this research details numerical finite element studies investigating the reflection of nonlinear ultrasonic waves within continuous functionally graded materials. The work directly addresses structural health monitoring challenges in nuclear and thermal power plant pressure vessels, where early-stage microscale degradation accumulates as a spatial function of depth. By updating linear and nonlinear constitutive relations to map exponential parameter variations, density shifts, and stiffness losses, the computational experiments successfully resolve complex wave field reflections. The results demonstrate that reflected higher longitudinal harmonics (\(2f, 3f, 4f, \dots\)) exhibit significantly higher energy profiles from fixed boundaries than from free boundaries and reveal that high-frequency signatures are fundamentally sensitive to spatial property distributions.

Source: https://doi.org/10.1121/10.0016325
https://pubs.aip.org/asa/jasa/article/152/4_Supplement/A294/2840281/Finite-element-studies-to-understand-reflection-of

- **The Journal of the Acoustical Society of America (JASA) Article on Direction-Independent Metamaterials and Multi-Objective Inverse Design – Pravinkumar Ghodake:** Published in the globally respected *Journal of the Acoustical Society of America*, this work establishes a rigorous optimization strategy for the inverse design of direction-independent elastic metamaterials engineered to control longitudinal and transverse waves independently. To support high-fidelity non-linear ultrasonic testings, the methodology solves transient forward finite element models as complex, multi-objective design problems using gradient-free algorithms. The mathematical framework successfully handles competing objectives: it suppresses parasitic, instrument-induced second harmonics (2*f*) without diminishing fundamental wave metrics, and employs min-max multi-objective optimization equations to minimize spectral tracking error while maximizing overall output pulse power.

  Source: https://doi.org/10.1121/10.0015924
  https://pubs.aip.org/asa/jasa/article/152/4_Supplement/A171/2839951/Inverse-design-of-direction-independent

- **INTER-NOISE Conference Proceedings on Inverse Design of Cylindrical Metamaterial Rods – Pravinkumar Ghodake:** Published officially via the Institute of Noise Control Engineering (INCE-USA) digital repository from the international *InterNoise* congress in Glasgow, this foundational research presents a transient shape optimization paradigm to engineer finite-length linear and nonlinear cylindrical metamaterial rods. Moving past traditional infinite-periodic assumptions that fail in physical laboratory setups due to pulse attenuation, the framework incorporates geometric and material nonlinearities to account for structural harmonic scattering. By driving transient finite element models with gradient-free routines like coordinate search and the Nelder-Mead algorithm, the approach systematically suppresses parasite instrument-induced 2nd harmonics while optimizing passband power for fundamental longitudinal pulse propagation modes.

Source: https://doi.org/10.3397/IN_2022_0667
https://ince.publisher.ingentaconnect.com/content/ince/incecp/2023/00000265/00000003/art00074

- **The Journal of the Acoustical Society of America (JASA) Article on Simultaneous Multi-Wave Metamaterial Regulation – Pravinkumar Ghodake:** Published in the premier journal of the Acoustical Society of America, this highly cited research breaks new ground by detailing the inverse design of linear periodic metamaterials capable of controlling both longitudinal and transverse waves simultaneously. Designing a single phononic device to manipulate multiple wave modes independently presents a historic challenge across elastodynamics. This framework solves the multi-wave inverse problem using transient finite element analysis, effective boundary conditions, and specialized optimization techniques. By introducing the discrete number of periodic cells (N) directly as an integer design parameter alongside structural layer widths, the system implements mathematical constraints that resolve complex mode conversions and cross-frequency mixing. Utilizing a gradient-free constrained optimization algorithm, the framework successfully attenuates parasitic second harmonics and regulates multi-wave interactions within a few minutes.

  Source:
  https://pubs.aip.org/asa/jasa/article/151/4_Supplement/A130/2838594/Metamaterials-to-control-both-longitudinal-and

- **The Journal of the Acoustical Society of America (JASA) Article on Non-Periodic Nonlinear Metamaterials and Multi-Objective Optimization – Pravinkumar Ghodake:** Published in the premier journal of the Acoustical Society of America, this research advances acoustic filtering by implementing shape optimization to engineer "non-periodic" mechanical metamaterials. To closely mimic authentic experimental conditions, the forward modeling framework accounts for both the material and geometric nonlinearities present within layered phononic crystals under high-frequency Gaussian short-pulse excitations. The inverse design methodology resolves a highly complex multi-objective optimization challenge: it systematically suppresses instrument-induced 2nd harmonics while concurrently maximizing fundamental 1st harmonic transmission and

preserving the authentic Gaussian pulse shape profile. This demonstrates profound applicability for precise wave manipulation across both non-destructive ultrasonic evaluations and large-scale seismic mitigation configurations.

Source: https://doi.org/10.1121/10.0010590 https://pubs.aip.org/asa/jasa/article/151/4_Supplement/A40/2839180/Inverse-design-of-non-periodic-nonlinear

- **The Journal of the Acoustical Society of America (JASA) Article on Cylindrical Metamaterial Rods and Shape Optimization – Pravinkumar Ghodake:** Published in the premier journal of the Acoustical Society of America, this highly impactful study introduces a robust shape optimization framework using a non-gradient algorithm and finite element modeling to design cylindrical metamaterials for wave control in cylindrical rods. By targeting single and multi-objective functions to suppress axial and radial displacements across end surfaces from monochromatic Gaussian pulses, the methodology determines optimal widths for layered elastic materials arranged periodically along subsurfaces and outer surfaces. The model integrates weak total-length constraints to ensure that every computationally generated design iteration remains readily manufacturable without requiring prior bandgap mapping, offering a far more computationally efficient and practical alternative to traditional topology optimization approaches.

  Source: https://pubs.aip.org/asa/jasa/article/151/4_Supplement/A56/2839293/Design-optimization-of-metamaterials-to-control

- **DEStech Transactions on Structural Health Monitoring Proceedings (Stanford University) – Pravinkumar Ghodake:** Published in the proceedings of the prestigious *International Workshop on Structural Health Monitoring (IWSHM)* at Stanford University, this research models the complex interaction of ultrasonic wave fields with randomly distributed, localized microstructural material damage. Moving beyond idealized homogeneous models, the study discretizes a 1D spatial domain as a long chain of spring-mass elements featuring random insertions of non-linear structural constraints. By independently analyzing symmetric (Bouc-Wen) and asymmetric (Generalized Bouc-Wen) tracking formulations under single-frequency and two-wave mixing sweeps, the computational framework demonstrates that random symmetric fields yield exclusively

odd harmonics, whereas asymmetric clusters activate a mixed spectral distribution of both odd and even components alongside extensive high-frequency structural energy transfers.

Source: https://www.dpi-proceedings.com/index.php/shm2021/article/view/36359

- **DEStech Transactions on Structural Health Monitoring Proceedings (Stanford University) – Pravinkumar Ghodake:** Published in the proceedings of the prestigious *International Workshop on Structural Health Monitoring (IWSHM)* at Stanford University, this research introduces an advanced finite element method (FEM) framework investigating the harmonic scattering of Shear Horizontal (SH) waves from highly localized material damages (such as local plasticity and shear bands). To bridge the gap where low-amplitude backscattered waves hinder experimental verification, this computational study maps wave actions across defects ranging from 0 to 20 mm. The models demonstrate a unique physical footprint: forward-scattered SH wave fields exhibit strictly odd harmonics, whereas both forward- and backscattered mode-converted Lamb waves reveal distinct even harmonic spectral signatures and cumulative static fields. The paper details how co-directional mixing zone expansions and two-way mixing sweeps trigger zero-group-velocity Lamb wave states directly inside localized defects, introducing a predictable baseline for solving non-linear acoustic inverse problems to quantify structural flaws.

  Source: https://www.dpi-proceedings.com/index.php/shm2021/article/view/36362

- **The Journal of the Acoustical Society of America (JASA) Article on Phononic Metamaterial Optimization – Pravinkumar Ghodake:** Published in the highly regarded *Journal of the Acoustical Society of America*, this work addresses a persistent challenge in non-destructive testing: the masking of pure material damage signatures by instrument-induced, system-generated higher harmonics. Moving past traditional trial-and-error design techniques, this research establishes a systematic size optimization problem incorporating realistic immersion boundary conditions and material geometric nonlinearities via COMSOL finite element modeling. By executing optimization sweeps through the Nelder-Mead algorithm under monochromatic and two-wave mixing pulse configurations, the engineered layered phononic crystals establish robust filtering stopbands. This methodology successfully suppresses unwanted second harmonic noise

amplitudes by up to 50 times when preserving exact output pulse shape profiles, and up to 80 times under unconstrained pulse shape optimization configurations.

Source: https://pubs.aip.org/asa/jasa/article/150/4_Supplement/A149/703713/Effective-design-of-metamaterial-to-stop-system

- **The Journal of the Acoustical Society of America (JASA) Article on Uncertainty Quantification and Bayesian Inversion – Pravinkumar Ghodake:** Published in the premier journal of the Acoustical Society of America, this research tackles parameter sensitivity and reliability challenges in mapping sub-surface flaws. Because the underlying wave equations dictate a highly non-linear forward problem, deterministic solutions are easily skewed by real-world experimental variations in input frequency or wave velocity. To resolve this instability, the study introduces an advanced uncertainty quantification framework integrating Reliability-Based Design Optimization (RBDO) and Bayesian Inverse (BI) methods. The computational experiments reveal that a Quantile Monte Carlo (QMC) approach produces far superior accuracy compared to standard first-order or index-based reliability methods. To circumvent heavy processing demands, the architecture couples these inverse methods with advanced Support Vector Machine Regression (SVR), Polynomial Chaos Expansions (PCE), and Kriging surrogate machine learning models—effectively collapsing computational runtime while preserving precision for non-destructive sizing applications.

  Source: https://pubs.aip.org/asa/jasa/article/150/4_Supplement/A92/706040/Uncertainty-quantification-of-quadratically

- **The Journal of the Acoustical Society of America (JASA) Article on A_0 Lamb Wave Harmonic Scattering – Pravinkumar Ghodake:** Published in the premier journal of the Acoustical Society of America, this research presents an extensive finite element method (FEM) investigation into the harmonic scattering of A_0 flexural Lamb waves from highly localized material damage zones. Moving past standard uniform degradation models, the study simulates discrete regions of micro-cracking and local plasticity in aluminum plates using the Murnaghan hyperelastic formulation while carefully eliminating linear impedance mismatch boundaries. The numerical configurations successfully isolate

displacement harmonics at the plate surface, demonstrating that for concise defect fields (5 to 20 mm), primary modes and second-harmonic signatures do not physically separate due to group velocity mismatches. Furthermore, by evaluating one-way and two-way two-wave guided mixing configurations with group velocity offsets, the research demonstrates complex sum, difference, and high-frequency mode conversions that are fundamentally sensitive to local defect scale and intensity.

Source: https://pubs.aip.org/asa/jasa/article/150/4_Supplement/A67/631761/Discussion-on-harmonic-scattering-of-A0-wave-from

- **Proceedings of the XXV International Congress of Theoretical and Applied Mechanics (ICTAM, Milan) – Pravinkumar Ghodake & Saurabh Biswas:** Published in the official proceedings of the prestigious International Union of Theoretical and Applied Mechanics congress, this co-authored research paper systematically evaluates one-dimensional, non-linear one-way two-wave mixing across rate-independent hysteretic mediums. Utilizing a discretized spring-mass chain integrated with parallel non-linear constraints, the investigation contrasts the classical scalar Bouc-Wen model against a multi-state friction-based hysteresis formulation. The simulation results demonstrate that while single-frequency inputs yield strictly odd harmonics, dual-frequency Gaussian pulse mixing accurately isolates sum and difference inter-modulation products. The paper uncovers distinct multi-frequency signatures, highlighting that the multi-state model successfully maps inner nested minor loops within the major structural loading path, bypassing the non-physical horizontal translations seen in scalar models to provide deep insights for non-destructive material damage characterization.

  Source:
  https://www.researchgate.net/profile/Pravinkumar_Ghodake/publication/354423054_Nonlinear_one-way_two-wave_mixing_in_a_media_with_rate_independent_hysteresis/links/62a8a3fe55273755ebeccd66/Nonlinear-one-way-two-wave-mixing-in-a-media-with-rate-independent-hysteresis.pdf

- **ASME Quantitative Nondestructive Evaluation Conference Proceedings on Pinched Hysteretic Materials – Pravinkumar Ghodake:** Published officially in the digital collection of the *American Society of Mechanical Engineers (ASME)* from the 48th Annual Review of Progress in QNDE, this research introduces an advanced numerical framework for analyzing nonlinear wave propagation in rate-independent pinched hysteretic elements. The 1D spatial domain is discretized as a long chain of spring-mass elements coupled in parallel with a pinched hysteresis formulation proposed by Biswas (2016). The computational experiments demonstrate that single-frequency waves generate exclusively odd higher harmonics within evolving pinched loops. Under dual-frequency one-way two-wave mixing sweeps, the inter-modulation response captures clear sum and difference frequencies distributed alongside tracking harmonics, mapping small, symmetric pinched tracing paths that bypass the standard non-physical loops seen in scalar models.

  Source: https://doi.org/10.1115/QNDE2021-74515
  https://asmedigitalcollection.asme.org/qnde/proceedings-abstract/QNDE2021/1131252

- **UKACM Conference Proceedings on Advanced Wave-Mixing and Acoustic Memory Mechanics – Pravinkumar Ghodake:** Published in the official proceedings archive of the *UK Association for Computational Mechanics (UKACM)* hosted by Loughborough University, this foundational research presents a novel computational paradigm for the quantification of early-stage fatigue degradation in metallic materials. Moving past standard single-pulse wave-mixing limits, the study utilizes the material's inherent **acoustic memory effect** by implementing two consecutive, time-delayed high-frequency Gaussian short pulses into a 1D spring-mass chain system modeled via Reid's pinched hysteresis formulation. By enforcing strict spatial resolution limits aligned with Courant-Friedrichs-Lewy (CFL) stability parameters, the numerical framework uncovers an unprecedented physical phenomenon where specific inter-modulation signatures (such as $3f_1 + f_2$, $f_1 + 3f_2$, and $3f_1 - f_2$) are uniquely activated via chronological time offsets. The paper demonstrates that tracking variations across the resulting asymmetric, translated pinched loops provides a powerful, computationally efficient avenue to solve complex non-linear ultrasonic inverse problems for structural health monitoring applications.

  Source:
  https://repository.lboro.ac.uk/articles/conference_contribution/Wave-

mixing_due_to_two_consecutive_short_pulses_in_Reid_s_pinched_hysteretic_material/14588388?file=27996309

- **The Journal of the Acoustical Society of America (JASA) Article on 1D Harmonic Scattering from Localized Hysteretic Damage – Pravinkumar Ghodake:** Published in the highly acclaimed *Journal of the Acoustical Society of America*, this work models the intricate wave mechanics of monochromatic ultrasonic wave fields interacting with localized, early-stage physical damage fields—such as plastic slip bands and microstructural fatigue cracking in metals. By discretizing a one-dimensional domain as a long spring-mass chain containing specialized parallel hysteretic elements at its center, the research evaluates the physical traits of symmetric and asymmetric hysteresis layouts. The computational experiments reveal that symmetric damage zones yield strictly odd harmonics, whereas asymmetric tracking models activate a concurrent distribution of both odd and even spectral components. The paper demonstrates that expanding the localized defect boundaries triggers a systematic, non-intuitive structural energy transfer that reduces fundamental forward-scattered fields while inducing cyclic increasing-and-decreasing interference patterns across the backscattered higher harmonic spectrum.

  Source:
  https://pubs.aip.org/asa/jasa/article/149/4_Supplement/A42/703279/One-dimensional-harmonic-scattering-from-a

- **The Journal of the Acoustical Society of America (JASA) Article on Functionally Graded Hysteretic Materials – Pravinkumar Ghodake:** Published in the distinguished *Journal of the Acoustical Society of America*, this study utilizes a customized 1D spring-mass chain configuration to analyze non-linear bulk wave propagation through a functionally graded symmetric hysteretic material model. The computational framework tracks complex harmonic generation signatures arising from the interplay between incoming single-frequency primary waves and their corresponding internal boundary reflections. By updating the spatial governing parameters, the numerical analysis captures the progressive deformation and structural layout shifts of the resulting dynamic hysteretic stress-displacement tracking curves to guide non-destructive material evaluation.

Source:
https://pubs.aip.org/asa/jasa/article/149/4_Supplement/A138/754501/Nonlinear-bulk-wave-in-a-functionally-graded

- **Elsevier Scopus Index / International Congress on Sound and Vibration Proceedings on Asymmetric Hysteretic Wave Propagation – Pravinkumar Ghodake:** Officially indexed in the global **Elsevier Scopus database** and published in the proceedings of the 27th International Congress on Sound and Vibration (ICSV 2021), this highly impactful research has achieved an **Field-Weighted Citation Impact (FWCI) of 1.51**, placing its scientific impact significantly above the global academic average. The publication marks a major milestone in non-linear elastodynamics by breaking a decades-old academic consensus. Traditional theoretical models (from 1994 to 2020) asserted that single-frequency ultrasonic wave paths through hysteretic media yield strictly odd harmonics. By discretizing a 1D spatial domain with advanced asymmetric rate-independent hysteretic elements based on the Generalized Bouc-Wen framework, this research demonstrates for the first time that asymmetric contact interfaces activate a continuous spectrum of **both odd and even higher harmonics** (2*f*, 3*f*, 4*f*, ...). Furthermore, by executing multi-frequency one-way two-wave mixing sweeps, the computational framework maps complex inter-modulation sum and difference frequency fields alongside nested minor tracing loops that physically intersect, delivering a novel baseline for non-destructive structural health monitoring and advanced acoustic characterization.

  Source: https://www.scopus.com/pages/publications/85117472334?origin=resultslist

- **Elsevier Scopus Index / ASME Quantitative Nondestructive Evaluation Proceedings on Pinched Hysteretic Materials – Pravinkumar Ghodake:** Officially indexed in the global **Elsevier Scopus database** and published via the *American Society of Mechanical Engineers (ASME)* from the 48th Annual Review of Progress in QNDE, this conference paper carries an impactful **Field-Weighted Citation Impact (FWCI) of 1.04**, establishing its scientific relevance above the global baselines. Backed by financial support from the **Acoustical Society of America (ASA) CIRE student grant**, this study investigates non-linear wave propagation in a 1D symmetric pinched rate-independent hysteretic material. By modeling parallel hysteretic spring-mass constraints using an improved

pinched hysteresis framework, the numerical solutions show that single-frequency waves generate exclusively odd higher harmonics within evolving pinched loops. Under dual-frequency one-way two-wave mixing sweeps, the inter-modulation response captures clear sum and difference frequencies distributed alongside tracking harmonics, mapping small, symmetric pinched tracing paths that surpass standard non-physical loops.

Source: https://www.scopus.com/pages/publications/85124142075?origin=resultslist

- **Elsevier Scopus Index / IWSHM Stanford University Proceedings on Localized SH Wave Harmonic Scattering – Pravinkumar Ghodake:** Officially indexed in the global **Elsevier Scopus database** and published within the proceedings of the *13th International Workshop on Structural Health Monitoring* (IWSHM) at Stanford University, this research introduces a finite element framework investigating high-frequency acoustic footprints in microscale fatigue zones. Backed by financial support from the **Acoustical Society of America (ASA) CIRE student grant** alongside an Indian Institute of Technology Bombay research fellowship, the study maps Shear Horizontal (SH) wave fields interacting with localized defect regions. The numerical simulations discover an explicit, predictable scattering phenomenon: forward-scattered SH wave paths yield strictly odd harmonics, whereas mode-converted backscattered and forward-scattered Lamb waves display exclusively even higher harmonics and cumulative static profiles. The paper details how co-directional and two-way mixing zones trigger zero-group-velocity Lamb wave states within localized imperfections, providing a foundational baseline for solving inverse optimization problems to isolate flaw sizes, intensities, and coordinates.

  Source: https://www.scopus.com/pages/publications/85139252220?origin=resultslist

- **Elsevier Scopus Index / IWSHM Stanford University Proceedings on Randomly Distributed Hysteretic Nonlinearity – Pravinkumar Ghodake:** Officially indexed in the global **Elsevier Scopus database** and published within the proceedings of the *13th International Workshop on Structural Health Monitoring* (IWSHM) at Stanford University, this conference paper outlines an advanced computational framework to map ultrasonic interactions inside disordered material zones. Fully backed by the **Acoustical Society of America (ASA) CIRE student grant** alongside an Indian Institute of Technology Bombay

research fellowship, the study models localized microstructural defects using a 1D spring-mass chain with randomized hysteretic insertions. By deploying parallel processing loops to independently solve classic symmetric (Bouc-Wen) and highly complex asymmetric (Generalized Bouc-Wen) mechanical formulations, the research maps how random structural distributions filter and scatter guided acoustic energies. The results demonstrate that while random symmetric networks trigger strictly odd spectral harmonics, asymmetric defect states break tracking constraints to activate a concurrent distribution of **both odd and even higher harmonics**, providing a critical diagnostic baseline for next-generation cyber-physical nondestructive evaluations.

Source: https://www.scopus.com/pages/publications/85139263071?origin=resultslist

- **Elsevier Scopus Index / Inter-Noise International Congress Proceedings on Cylindrical Metamaterial Rods – Pravinkumar Ghodake:** Officially indexed in the prestigious **Elsevier Scopus database** and published within the proceedings of the *51st International Congress and Exposition on Noise Control Engineering* (Inter-Noise 2022, Glasgow), this research introduces a robust shape optimization framework for finite-length linear and nonlinear wave-guiding rods. Fully backed by the **Acoustical Society of America (ASA) student grant** and an Indian Institute of Technology Bombay institute scholarship, the study addresses the real-world manufacturing constraints where classical infinite-periodic lattice idealizations break down due to short-pulse decay. By coupling transient finite element modeling directly with gradient-free numerical tracking loops (Coordinate Search and the Nelder-Mead algorithm), the computational setup treats concurrent material and geometric nonlinearities to mitigate parasitic harmonic scattering. The optimized layered structures successfully filter out system-generated 2nd noise harmonics while maximizing transmission across fundamental longitudinal wave paths.

  Source: https://www.scopus.com/pages/publications/85147449847?origin=resultslist

- **Elsevier Scopus Index / META International Conference Proceedings on Direction-Dependent Metamaterial Synthesis – Pravinkumar Ghodake:** Officially indexed in the global **Elsevier Scopus database** and published within the proceedings of the *12th International Conference on Metamaterials, Photonic Crystals and Plasmonics* (META

2022, Torremolinos, Spain), this research details a milestone in non-linear elastodynamics by introducing an advanced inverse design framework for asymmetric wave-guiding systems. Financed by the **Acoustical Society of America student grant** and an Indian Institute of Technology Bombay scholarship, the methodology constructs time-dependent inverse problems to optimize a direction-dependent mechanical metamaterial. Operating independently and concurrently in both time and frequency domains, the engineered structure successfully filters parasitic second harmonics ($2f = 4$ MHz) in one direction while maximizing the fundamental transmission power ($f = 2$ MHz) of a short Gaussian pulse moving in the exact opposite direction.

Source: https://www.scopus.com/pages/publications/85179871283?origin=resultslist

- **Elsevier Scopus Index / META International Conference Proceedings on Functionally Graded Metamaterial Unit Cells – Pravinkumar Ghodake:** Officially indexed in the global **Elsevier Scopus database** and published within the proceedings of the *13th International Conference on Metamaterials, Photonic Crystals and Plasmonics* (META 2023, Paris, France), this publication explores advanced wave manipulation mechanics using a novel architectural building block. The research conducts multiple computational experiments analyzing a non-linear, functionally graded metamaterial unit cell sandwiched symmetrically between linear elastic media of matching acoustic impedance. By mapping the sensitivities of parameter configurations under both forward and inverse design approaches, the modeling framework demonstrates how the continuous spatial distribution of non-linear properties can be tailored to precisely structure, trap, or filter the harmonic responses of high-frequency propagating elastic waves.

  Source: https://www.scopus.com/pages/publications/85174567581?origin=resultslist

- **Elsevier Scopus Index / META International Conference Proceedings on Time-Varying Elastic Metamaterials – Pravinkumar Ghodake:** Officially indexed in the global **Elsevier Scopus database** and published within the proceedings of the *13th International Conference on Metamaterials, Photonic Crystals and Plasmonics* (META 2023, Paris, France), this research documents extensive computational experiments designed to investigate the emerging non-linear response signatures produced by linear,

time-dependent elastic metamaterials. By modulating structural material parameters dynamically across the temporal domain, the research maps advanced wave manipulation capabilities. These include the automated generation of higher harmonics, precise control over scattered spectral components, and the regulation of mixed-frequency combinations during single-frequency monochromatic wave interactions and co-directional two-wave guided mixing sequences.

Source: https://www.scopus.com/pages/publications/85174568088?origin=resultslist

- **The Journal of the Acoustical Society of America (JASA) Article on Damage Sizing Inverse Problems – Pravinkumar Ghodake & Salil S. Kulkarni:** Published in the premier journal of the Acoustical Society of America, this co-authored research establishes a mathematical foundation for sizing localized material degradation zones. The study models non-uniform macro-defects as discrete regions exhibiting distinct quadratic or cubic elastic nonlinearities, sandwiched between linear matrix zones. By tracking the transmission and reflection fields of propagating waves, the framework maps how a highly non-linear forward problem creates multiple non-unique solutions. To resolve this ambiguity, the methodology reformulates the parameter estimation as a constrained optimization problem solved via gradient and non-gradient search algorithms. Verified with noise-corrupted synthetic data, the research reveals that while quadratically non-linear defects require at least two independent multi-frequency sweeps to isolate accurately, a single high-frequency experiment is sufficient to reliably quantify cubically non-linear structural damage sizes.

  Source:
  https://pubs.aip.org/asa/jasa/article/148/4_Supplement/2459/642916/Damage-size-quantification-in-one-dimensional-bar

- **The Journal of the Acoustical Society of America (JASA) Article on Hysteretic Wave Propagation – Pravinkumar Ghodake & Saurabh Biswas:** Published in the premier journal of the Acoustical Society of America, this co-authored research presents a detailed numerical investigation into high-frequency wave mechanics within rate-independent hysteretic media. By modeling an aluminum bar subjected to low-cycle fatigue as a discrete

1D spring-mass chain, the study provides a critical comparative analysis between the classical scalar Bouc-Wen model and the more advanced Two-State friction-based hysteresis formulation. Spatially resolved to capture delicate high-frequency distortions, the computational framework demonstrates that while both models track the activation of dominant odd higher harmonics, the Bouc-Wen model is physically constrained and fails to resolve localized minor loops caused by partial reversals. Conversely, the Two-State model successfully captures fine inner minor loop tracking trajectories, offering a highly accurate numerical baseline for characterising realistic microstructural contact damage.

Source: https://pubs.aip.org/asa/jasa/article/148/4_Supplement/2693/645287/Wave-propagation-in-a-one-dimensional-bar-with

- **Acoustical Society of America (ASA) Student Summer Talks Presentation on Topology Optimization and Mechanical Filters – Pravinkumar Ghodake:** Featured in the official proceedings schedule of the *Acoustical Society of America (ASA) Student Summer Talks*, this research presents a structural optimization framework developed to eliminate parasitic, instrument-induced higher harmonics that corrupt diagnostic wave measurements during non-destructive evaluations. To isolate pure material damage signatures, the methodology outlines the inverse design of a localized mechanical filter consisting of alternating, bonded disks of differing materials and thicknesses but sharing a constant cross-sectional area. The filter is evaluated using one-dimensional wave analysis and mathematically optimized using thickness dimensions as design variables to establish robust bandgaps that inject clean monochromatic waves into a testing specimen.

  Source: https://asastudents.org/wp-content/uploads/2020/07/Virtual-ASA-Student-Summer-Talks-Schedule-v02.pdf

- **NASA ADS Index / APS March Meeting Presentation on Nonlinear Wave Cloaking and Elastic Invisibility – Pravinkumar Ghodake:** Indexed within the highly authoritative NASA Astrophysics Data System (ADS) and presented at the APS March Meeting 2023, this publication introduces a time-dependent constrained optimization framework developed to achieve true **nonlinear cloaking** and absolute **invisibility** against longitudinal ultrasonic wave fields. To overcome the challenges of modeling complex

arrays of incipient microstructural flaws (such as dislocation networks and micro-voids), the framework constructs a parametric framework modeling clusters of elliptical hyperelastic inclusions (Murnaghan material) embedded in a linear matrix. By utilizing the Nelder-Mead gradient-free optimization algorithm on transient finite element models, the setup coordinates inclusion dimensions, shape profiles, rotation angles, and relative spatial positions. The resulting optimal structural cluster configuration achieves total self-cancellation of transmitted higher harmonics, rendering complex fatigue-damaged zones completely elastically invisible to incoming longitudinal diagnostic waves.

Source: https://ui.adsabs.harvard.edu/abs/2023APS..MARJJ8009G/abstract

- **Isaac Newton Institute (INI) Canonical Scattering Summit Workshop – Pravinkumar Ghodake:** Hosted at the world-renowned Isaac Newton Institute for Mathematical Sciences at the University of Cambridge, this highly competitive, peer-reviewed international research workshop gathers global leaders in diffraction theory, metamaterial design, and advanced elastodynamics. Selected as an elite participant representing the Indian Institute of Technology, this research engagement focused on the mathematical and computational modeling of advanced spatial variations in composite wave media. The workshop showcased state-of-the-art developments in matrix Riemann-Hilbert formulations, high-frequency boundary scattering, and non-gradient optimization paradigms, directly reinforcing the structural implementation of **nonlinear wave cloaking** and acoustic invisibility matrices in finite, multi-layered mechanical filtering lattices.

  Source: https://www.newton.ac.uk/event/mwsw01/

- **Metamaterials Congress International Peer-Review Selection on Periodic and Quasi-Periodic Systems – Pravinkumar Ghodake:** This solo-authored technical contribution successfully cleared the rigorous, elite selection process for the *16th International Congress on Artificial Materials for Waves and Electrons* (Metamaterials 2022). Representing one of the exceptionally few selected authors from India, this research establishes an advanced inverse-design optimization strategy for periodic and quasi-periodic non-linear mechanical metamaterials. By setting up a forward finite element scheme integrated with transient

Nelder-Mead search loops, the platform automates the geometric distribution of elastic layers to suppress unwanted high-frequency harmonics. Though cleared for presentation alongside plenary sessions from global pioneers, the paper was formally withdrawn due to institutional travel funding constraints.

Source:
https://congress2022.metamorphose-vi.org/index.php/component/svisor/?task=showScheduleSvisor&Itemid=156

- **The Journal of the Acoustical Society of America (JASA) Official Technical Board Appointments – Pravinkumar Ghodake:** Formally published in the distinguished *Journal of the Acoustical Society of America*, this official record recognizes senior scientific appointment to three prestigious national Technical Committees of the Acoustical Society of America (ASA): **Computational Acoustics** (Term to 2025), **Engineering Acoustics** (Term to 2025), and **Education in Acoustics** (Term to 2025). Appointed alongside leading international authorities in wave mechanics, this high-stature governance designation chronicles sustained contributions to the fields of advanced numerical simulation, non-destructive testing platforms, and international academic outreach within the global acoustics and physics community.

  Source: https://pubs.aip.org/asa/jasa/article/152/3/1970/2839575/Acoustical-News

- **The Journal of the Acoustical Society of America (JASA) Multi-Committee Board Appointments and Executive Council Leadership – Pravinkumar Ghodake:** Formally published in the premier peer-reviewed journal *The Journal of the Acoustical Society of America*, this official society directory recognizes extended senior appointment and high-profile leadership roles across five distinguished national boards of the Acoustical Society of America (ASA). The comprehensive governance listing records simultaneous terms on the **Computational Acoustics** Technical Committee (to 2025), **Engineering Acoustics** Technical Committee (to 2025), **Physical Acoustics** Technical Committee (to 2026), **Structural Acoustics and Vibration** Technical Committee (to 2026), and the **Education in Acoustics** Committee (to 2025). Concurrently, this national administrative record chronicles selection as the executive **Student Council**

Representative for Computational Acoustics (through Spring 2024). These consecutive administrative and technical selections alongside top-tier international authorities underscore a globally recognized tenure in shaping advanced numerical benchmarking, wave propagation physics, structural optimization paradigms, and academic student mentorship within the international physics community.

Source: https://doi.org/10.1121/10.0020860
https://acousticalsociety.org/wp-content/uploads/2020/01/1539_1_10.0020860.pdf

- **Figshare Open Science Framework Dataset Directory – Pravinkumar Ghodake:** Institutional data repository housing public dataset materials, computational files, and teaching assistantship metric tracks.

  Source: https://figshare.com/authors/Pravinkumar_Ghodake/3701242

📋 Official Google Sites Directory

Section Description: This directory serves as a centralized, unedited map of my official Google Sites pages. It provides a structured index of my academic records, research outputs, and personal timelines to ensure accurate indexing across global search engines.

🎓 Core Academic & Research Portal Links

- **Official Homepage:** Central landing page mapping research in metamaterials, nonlinear waves, and inverse design problems.

  Source: https://sites.google.com/view/pravinkumarghodake/home

- **Research Biography:** Detailed independent background in computational, theoretical, and experimental solid mechanics.

  Source: https://sites.google.com/view/pravinkumarghodake/biography

- **Research Interests:** Comprehensive breakdown of fields including guided waves, nonlinear ultrasonics, and structural health monitoring.

  Source: https://sites.google.com/view/pravinkumarghodake/research-interests

- **Research Outputs:** Archive of academic publications, unit cell proposals, invited conference talks, and experimental media.

  Source: https://sites.google.com/view/pravinkumarghodake/research

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Academic, Professional & Site Records Records & Contact Links

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- **Teaching Record:** Teaching profile documenting instructional activities and responsibilities across WCE Sangli, IIT Kanpur, and IIT Bombay.

  Source: https://sites.google.com/view/pravinkumarghodake/teaching

- **Completed Courses:** Coursework registry documenting completed coursework in areas including continuum mechanics and machine learning.

  Source: https://sites.google.com/view/pravinkumarghodake/courses

- **Awards & Distinctions:** Awards & Legacy – Acknowledgments, honors, and core personal philosophy.

  Source: https://sites.google.com/view/pravinkumarghodake/awards

- **Direct Connect:** Contact directory containing professional contact information and related communication channels.

  Source: https://sites.google.com/view/pravinkumarghodake/connect

- **Miscellaneous Hub:** Miscellaneous Index – Gateway directory tracking numerical code bases, public commentary, and external presentations.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous

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📜 Creative Archives & Public Logs

- **Snapshots Blog:** Snapshots Blog – A public timeline detailing personal experiences, short blogs, and independent researcher insights.

  Source: https://sites.google.com/view/pravinkumarghodake/snapshots

- **Creative Entry Log 0:** Creative Archive 0 – Initial creative entries, personal milestones, and side projects.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-0

- **Creative Entry Log I:** Creative Archive I – Hobbies journal and reflective personal entry tracking.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-i

- **Creative Entry Log II:** Creative Archive II – Multi-domain interest tracking and personal commentary log.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-ii

- **Creative Entry Log III:** Creative Archive III – Ongoing personal development log and independent journey archive.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-iii

- **Creative Entry Log IV:** Creative Archive IV – Media tracking and creative independent writing entry.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-iv

- **Creative Entry Log V:** Creative Archive V – Independent thought logging and personal creative outlet log.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-v

- **Creative Entry Log VI:** Creative Archive VI – Ongoing milestone tracking log and personal journey notes.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-vi

- **Creative Entry Log VII:** Creative Archive VII – Independent personal expression index and analytical logging.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-vii

- **Creative Entry Log VIII:** Creative Archive VIII – Creative tracking module and independent researcher personal timeline.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-viii

- **Creative Entry Log IX:** Creative Archive IX – Continuous creative narrative mapping personal observations.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-ix

- **Creative Entry Log X:** Creative Archive X – Milestone diary tracking creative pursuits and independent observations.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-x

- **Creative Entry Log XI:** Creative Archive XI – Reflective thoughts, philosophical framework, and independent logs.

Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-xi

- **Creative Entry Log XII:** Creative Archive XII – Creative milestone logger and multi-disciplinary activity tracker.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-xii

- **Creative Entry Log XIII:** Creative Archive XIII – Deep observation diary and personal milestone logging.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-xiii

- **Creative Entry Log XIV:** Creative Archive XIV – Strategic thought archive, creative pieces, and personal data.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-xiv

- **Creative Entry Log XV:** Creative Archive XV – Core timeline check-in, creative logging, and independent milestones.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-xv

- **Creative Entry Log XVI:** Creative Archive XVI – Personal repository for narrative prose and individual insights.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-xvi

- **Creative Entry Log XVII:** Creative Archive XVII – Independent progress tracker and narrative logging data.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-xvii

- **Creative Entry Log XVIII:** Creative Archive XVIII – Analytical creative log charting continuous personal growth.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-xviii

- **Creative Entry Log XIX:** Creative Archive XIX – Creative reflections module and individual milestone verification.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-xix

- **Creative Entry Log XX:** Creative Archive XX – Strategic creative outlet tracking independent research journeys.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-xx

- **Creative Entry Log XXI:** Creative Archive XXI – Narrative logs, life observations, and secondary interests checklist.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-xxi

- **Creative Entry Log XXII:** Creative Archive XXII – Creative archive containing personal entries and milestone records.

  Source: https://sites.google.com/view/pravinkumarghodake/miscellaneous/hobbies-xxii

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💼 Professional & Social Media Anchors

- **LinkedIn:** Comprehensive professional history, education tracking, and professional network.

  Source: https://www.linkedin.com/in/pravinkumarghodake

- **X (formerly Twitter):** Official X Profile | @pravinkumar_727– Real-time public statements, social updates, and verified broadcasts.

  Source: https://x.com/pravinkumar_727

- **YouTube:** Archive of technical presentations and travel recordings.

  Source: https://www.youtube.com/channel/UCDZAnjmqN0Geh_0h3IORp0g/videos

- **Instagram:** Personal photo log.

  Source: https://www.instagram.com/pravinkumarghodake

- **Facebook:** Personal profile.

  Source: https://www.facebook.com/pravinkumar.ghodake.07